Loan Default Prediction

A Data-Driven Credit-Risk Model for Home Equity Line of Credit Underwriting

Capstone Project — Practical Data Science (Full-Code)
MIT Professional Education — Applied AI & Data Science Program | May 2026 Cohort

Presented by: Divya Yadav

The Business Problem

Why this matters to the bank

Traditional underwriting can be slower and less consistent at scale

Interest income is an important source of retail-bank revenue

Loan defaults reduce interest income and increase credit losses

Credit decisions require explainable and defensible reasons for adverse actions

OBJECTIVE

"Predict borrowers at higher risk of default and identify the key factors that should inform credit-approval decisions."

— Official Great Learning Problem Statement

19.95%

historical default rate
(1,189 of 5,960 applicants)

⭐ **Key Takeaway: The goal is to identify high-risk borrowers earlier and support more consistent, explainable credit decisions.**

The Data

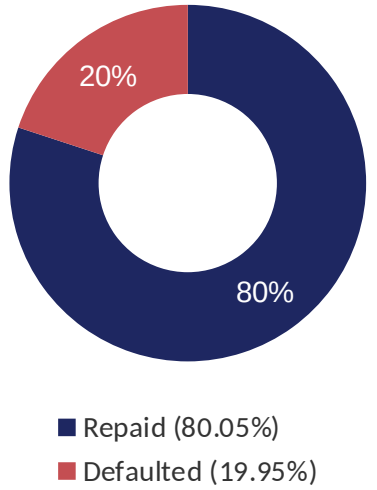
HMEQ: Home Equity applicant records

5,960	12	0	21.3%
Applicants' records	Input variables	Duplicate records	DEBTINC missing values

Feature groups

- **Loan/property:** LOAN, MORTDUE, VALUE
- **Credit history:** DEROG, DELINQ, CLAGE, NINQ, CLNO
- **Affordability:** DEBTINC (debt-to-income)
- **Applicant profile:** REASON, JOB, YOJ

Target: BAD



1 = defaulted / severely delinquent
0 = loan repaid

⭐ **Key Takeaway: The dataset captures loan, credit-history, affordability, and applicant characteristics, with about 1 in 5 records showing default.**

The Most Important EDA Finding

Missingness is strongly associated with default

62.0%

default rate when DEBTINC (debt-to-income) is MISSING

(1,267 applicants — 21.3% of the dataset)

8.6%

default rate when DEBTINC IS reported

(4,693 applicants — 78.7% of the dataset)

This missingness indicator became the strongest feature in the final model — turning a data-quality issue into a meaningful risk signal.

⭐ **Key Takeaway: Applicants with missing DEBTINC had a 7.2× higher historical default rate, making missingness an important risk signal to investigate.**

Methodology

Leakage-safe pipeline, four required model families

1

Split first

Stratified 70/30 train-test split

All preprocessing statistics learned from TRAIN only

2

Flag & impute

Flag missing values, then impute

Missingness indicators added; median/mode values learned from TRAIN only

3

Encode & scale

Transform features

One-hot encode categorical variables; cap outliers and scale where required

4

Model & tune

Build, compare & tune models

Four model families evaluated with class weighting and cross-validated hyperparameter tuning

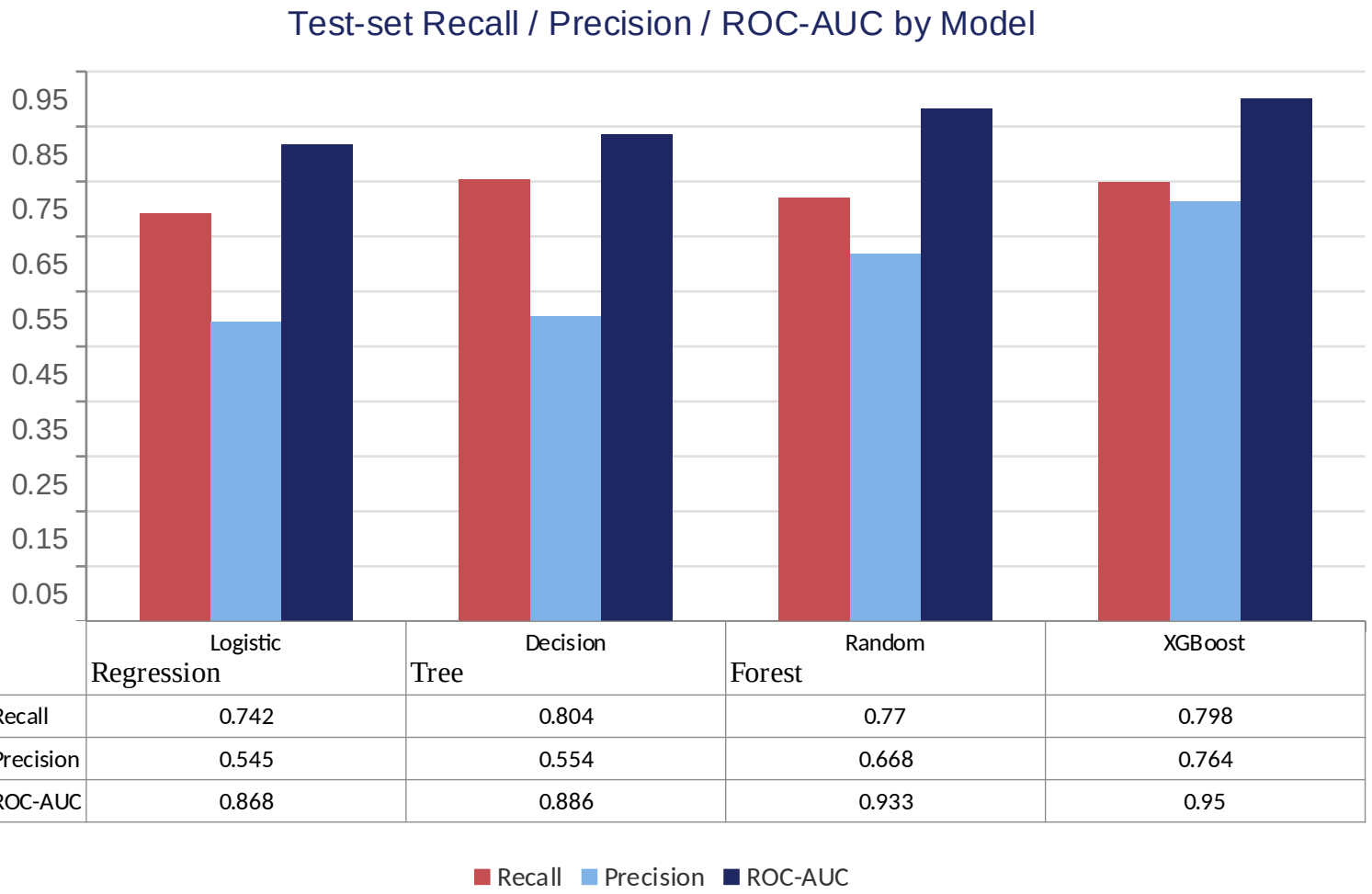
Why Recall, not Accuracy

Missing a true defaulter can expose the bank to credit loss, while incorrectly flagging a good applicant can be addressed through human review. Therefore, Recall is the primary metric, with Precision and F1 used to assess the trade-off.

⭐ **Key Takeaway: The pipeline prevents data leakage and prioritizes identifying as many true defaulters as possible while controlling unnecessary reviews.**

Model Comparison

Four tuned models compared on unseen test data



	Logistic Regression	Decision Tree	Random Forest	XGBoost
Recall	0.742	0.804	0.77	0.798
Precision	0.545	0.554	0.668	0.764
ROC-AUC	0.868	0.886	0.933	0.95

XGBoost (Tuned) provides the strongest overall performance

Recall	**0.798**
Precision	**0.764**
F1	**0.781**
ROC-AUC	**0.950**

Not chosen for ROC-AUC alone — Random Forest was close (0.933). XGBoost won on Recall + Precision + F1 together.

⭐ **Key Takeaway: Tuned XGBoost provides the best balance—identifying 79.8% of actual defaulters while 76.4% of high-risk predictions were actual defaulters.**

Handling Class Imbalance & Threshold

80/20 split handled via class weighting + threshold, not SMOTE

Why not SMOTE?

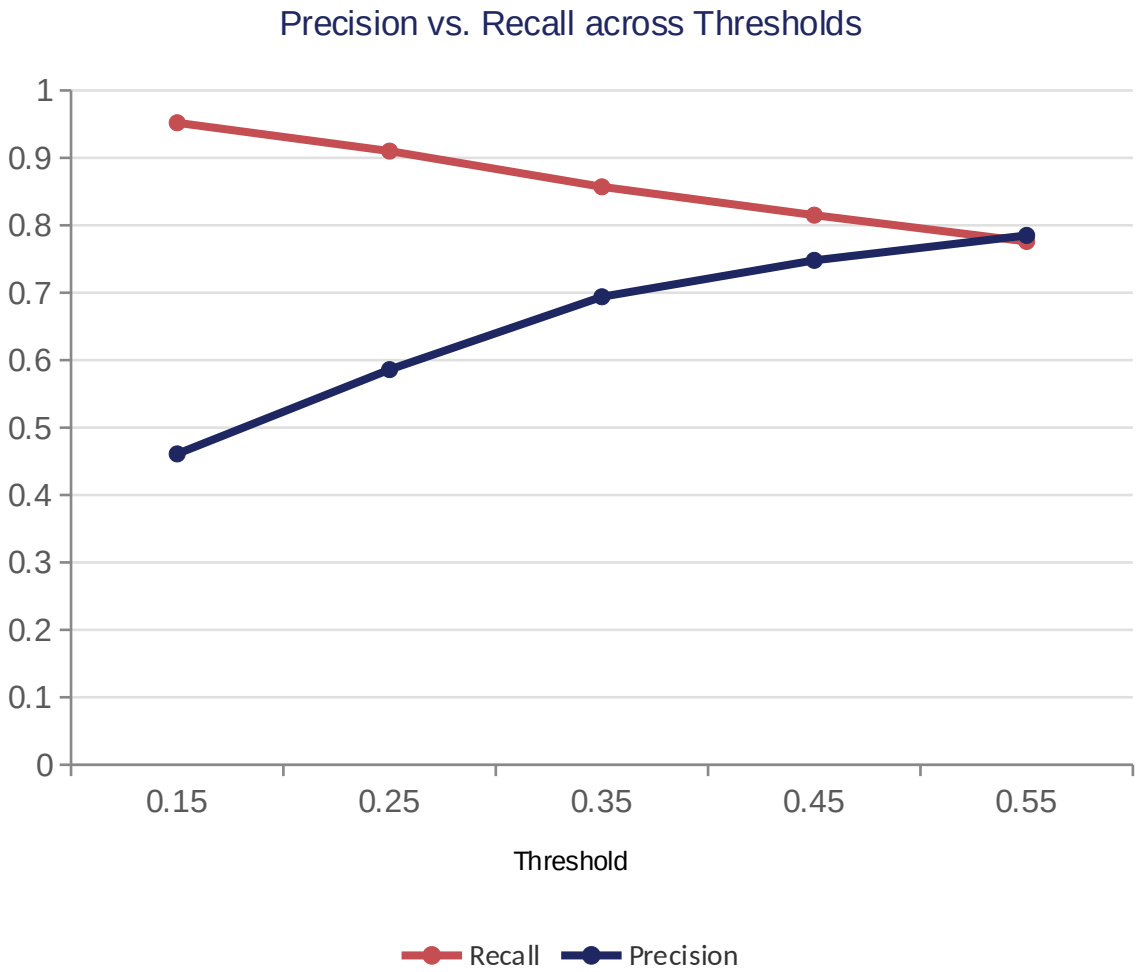
At ~4:1, this is moderate imbalance, not extreme

Class weighting (class_weight / scale_pos_weight) directly penalizes missing a defaulter during training

Avoids the risk of overfitting to synthetic minority-class points

Moving the decision threshold (XGBoost)

0.50 is a convention, not a business decision. Lowering it trades Precision for Recall. allowing the bank to catch more potential defaulters.



⭐ **Key Takeaway: Lowering the threshold from 0.50 to 0.30–0.40 can catch more defaulters but increases the number of applicants sent for review.**

Final Model Recommendation

XGBoost (tuned) + Logistic Regression (transparent companion)

Hyperparameters (RandomizedSearchCV)

```
n_estimators = 300
max_depth = 4
learning_rate = 0.05
subsample = 0.9, colsample_bytree = 0.9
scale_pos_weight ≈ 4.01 (class-imbalance correction)
```

Recommended operating threshold: 0.30–0.40

Selected using cross-validated training probabilities and confirmed on the untouched test set.

Final threshold should be validated against the bank's actual cost of missed defaults and manual review.

Output = risk-ranking score, not calibrated probability.

**Deployed as decision support —
not an automated approve/decline switch**

Human underwriter retains final say

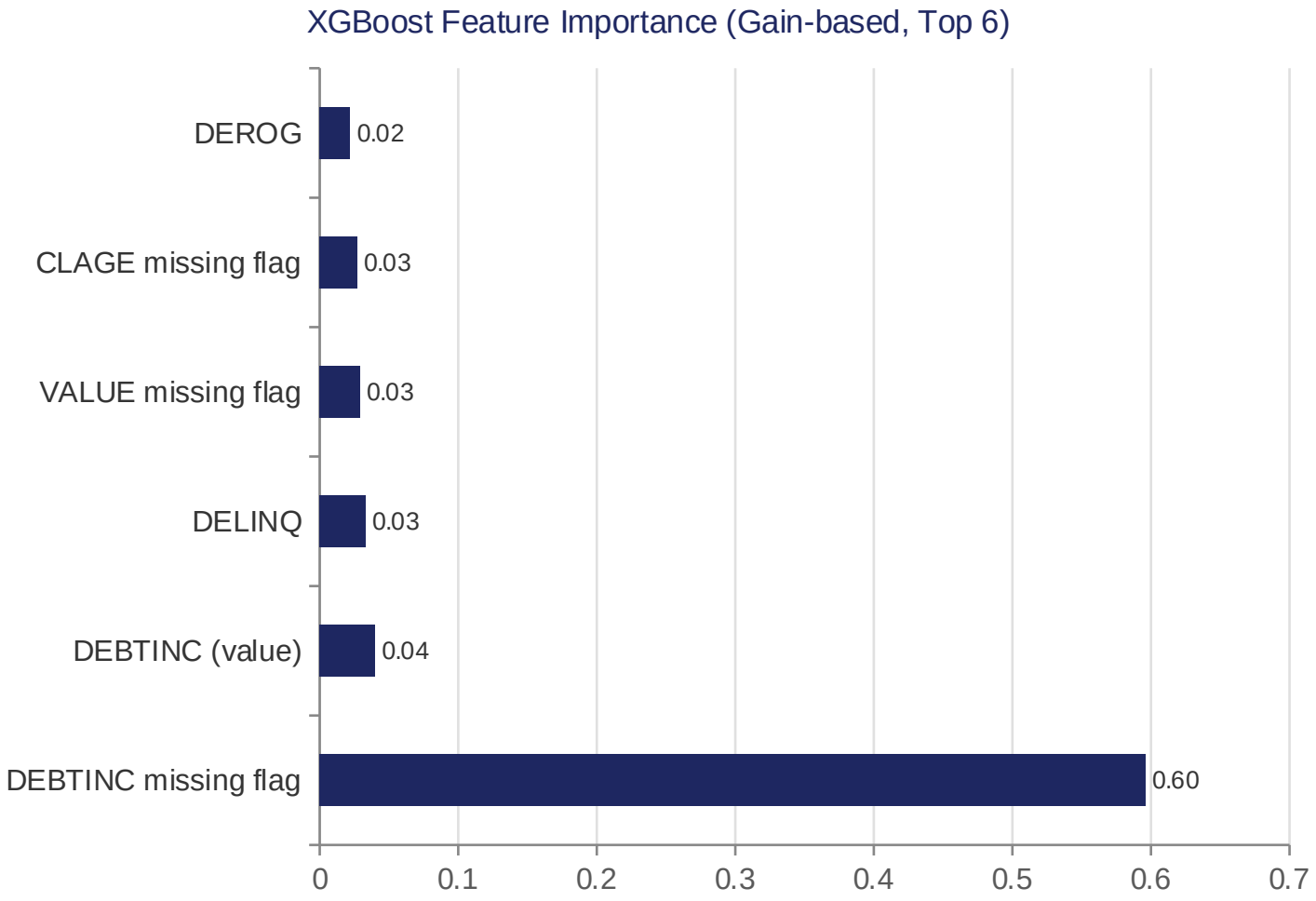
LogReg coefficients + feature importance support a defensible adverse-action rationale, pending formal compliance validation

Shadow-mode pilot before live use

⭐ **Key Takeaway: Tuned XGBoost is recommended for prediction, supported by Logistic Regression for interpretability and human review for final decisions.**

Key Risk Drivers

Triangulated across 3 independent explainability methods



What this means for underwriting

Debt-to-income reporting status is the dominant driver, followed by credit-history variables and missing-data indicators.

Confirmed across gain importance, permutation importance, AND Logistic Regression coefficients

Association, not proof of causation, a hypothesis for further investigation, not a settled mechanism

⭐ **Key Takeaway: Debt-to-income reporting status is the strongest risk signal, highlighting the importance of both credit information and data completeness.**

Business Recommendations

What each stakeholder should do next

Credit Risk	Validate the 0.30–0.40 threshold range against real loss-given-default and review-cost figures
Underwriting Ops	Pilot the model in shadow mode before any live decisioning use
Compliance / Fair Lending	Check the model for fairness across different groups before using it.
Data / IT	Investigate why DEBTINC is missing for 21% of applicants and improve data-capture processes

⭐ **Key Takeaway: Validate the model, protect fair-lending outcomes, and fix data-quality issues before production deployment.**

Risks, Limitations & What This Project Cannot Claim

Honest boundaries matter as much as the results

Selection bias: *the model was trained only on previously approved applicants, so it has not been tested on declined applicants.*

No demographic data: *we cannot claim the model is free from bias. A full fairness review would need additional data.*

No cost data: *the project provides a cost framework, but not estimated dollar benefits.*

XGBoost *is harder to explain than Logistic Regression, so additional explanation methods are needed.*

Model drift: *applicant behavior and economic conditions can change, so the model needs regular monitoring and retraining.*

⭐ **Key Takeaway: The model shows strong potential, but it needs further testing, monitoring, and validation before real-world use.**

Conclusion

A tuned XGBoost model identifies ~80% of actual defaulters at 76% precision — the best overall balance among the models tested.

Credit history and data completeness — especially DEBTINC availability — are the strongest risk signals in the data.

Use the model as decision support, with Logistic Regression for interpretation, human review, and fairness checks before deployment.

Thank you — questions welcome